

Where to Go Next: Minimax Site-Selection for External Validity

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Abstract

A multi-site experiment supports its external-validity claim through a cross-site moment identity $\Sigma\theta + b = 0$. When the number of sites S is small relative to the site-level covariate dimension, this identity is rank-deficient and leaves the target-site treatment effect only partially identified. In recent work, [Boyarsky, Egami, and Namkoong \[2026\]](#) formalize this partial identification and derive debiased bounds on $\psi(\theta; \gamma)$ at a user-chosen target profile (t_m, t_r) under analyst-specified affine restrictions $r(\theta) \leq 0$. In this work we study the *inverse* design question. Given a pool of candidate sites with known site-level context $(C_m^{(k)}, C_r^{(k)})$ but no experimental data, which site should an investigator go and run to most strengthen external-validity claims about a population of target profiles? We formulate the problem as a minimax design over the width of the partial-identification region. We establish a geometric characterization that, without restrictions, reduces to an “expand-the-affine-span” rule: the optimal candidate most reduces the worst-case distance from the target functional to the enlarged moment range, which under a balanced embedding is a weighted affine-hull expansion rule for the site-level context vectors, with convex-hull expansion the special case in which targets lie outside the existing affine span. Under compact affine restrictions, we derive a tractable surrogate. The criterion has the same three-regime structure as the leave-one-site-out diagnostic of [Boyarsky et al. \[2026\]](#): candidate sites either (i) strictly reduce the null space of Σ and can collapse unbounded widths to finite ones, (ii) translate the moment-implied affine subspace closer to the target profile at fixed rank, or (iii) restore feasibility of substantive restrictions that were incompatible with the existing pool. We also give sufficient conditions under which adding a site does not increase the worst-case width. We illustrate these findings on synthetic data. In our simulations, candidate sites that satisfy our conditions reduce the minimax width. Together, these results give practitioners a simple rule for choosing the next site to strengthen the external validity of experimental findings.

1 Introduction

A multi-site experiment supports its external-validity claim through the *cross-site* variation in site-level context. If the treatment effect is stable across sites that differ substantially in their moderating attributes, then it is credible that the effect will transport to a new site whose moderators lie within the observed range. Conversely, when the observed sites cover only a small part of the space of site-level moderators, the cross-site moment identity $\Sigma\theta + b = 0$ is rank-deficient. The target-site treatment effect at a moderator profile (t_m, t_r) outside the span of observed sites is then only partially identified, no matter how large the individual-level samples are [[Boyarsky, Egami, and Namkoong, 2026](#)].

An investigator facing this situation has two options. She may *tighten the identification set* by imposing additional substantive restrictions $r(\theta) \leq 0$ (effect-ordering, sign or monotonicity

constraints, norm bounds, etc.). This trades a sharper bound for additional untestable assumptions. Or she may *expand the evidence base* by running a new experiment in an additional site. The first option is the subject of [Boyarsky et al. \[2026\]](#). In this paper we address the second.

Specifically, we consider the situation in which the investigator has a small pool $\mathcal{K} = \{1, \dots, K\}$ of *candidate* sites. For each candidate site the investigator knows the site-level context $(C_m^{(k)}, C_r^{(k)})$ in advance, cheaply obtained from administrative data, public statistics, a pilot survey, or an established literature, but has no individual-level experimental data. Running the experiment in any one candidate site carries an investigator-specified cost c_k (monetary, logistical, ethical). The question is:

Given the existing pool of S sites and a range $\mathcal{T}$ of target profiles to which the investigator wishes to transport the effect, which candidate site $k^* \in \mathcal{K}$ most improves external-validity claims, in the sense of most shrinking the partial-identification width over $\mathcal{T}$?

We cast the problem as a **minimax design**:

$$k^* \in \arg \min_{k \in \mathcal{K}} \sup_{(t_m, t_r) \in \mathcal{T}} w_r(t_m, t_r; \Sigma_{S \cup \{k\}}),$$

where $w_r(t_m, t_r; \Sigma)$ is the width of the partial-identification interval under restriction r evaluated at target (t_m, t_r) . The key observation is that the width w_r depends on the candidate site through Σ , the *design matrix* of the moment condition, and Σ is a function of known site-level quantities (C_m, C_r) only. This is what makes the design problem tractable without running the candidate experiments. The right-hand side b would move once the experiment is run. The site-level context determines the rank and moment-range geometry, and therefore determines *unrestricted* finiteness of the width, conditional on feasibility. Under compact restrictions, however, b also matters: it translates the moment-implied affine slice through the restriction set and can change the width (Section 6.2). We therefore treat the context-driven geometry as the design-time object and evaluate candidates using either an outcome forecast or a robust uncertainty set for Δb_k (Section 5.3).

Contributions. (1) We formalize the site-selection problem in the partial-identification framework of [Boyarsky et al. \[2026\]](#). We show that the partial-identification width depends on the candidate through a low-rank update of Σ (rank at most two) that is known before the experiment is run. (2) We establish that, in the unrestricted limit $r \equiv 0$ with mild regularization, the minimax rule selects the candidate whose addition most reduces the worst-case distance from the target functional $c(t)$ to the enlarged moment range (Proposition 3). Under a balanced isometric embedding this becomes a weighted *affine-span* expansion rule in site-context space (Corollary 2), with the weighting determined by $\mathcal{T}$. Convex-hull expansion is the right design intuition exactly when the target set lies outside the existing affine span in target-relevant directions. (3) Under compact affine restrictions $r(\theta) \leq 0$, we derive a tractable surrogate, a finite LP over the candidate pool, that has the same three-regime structure as the leave-one-site-out diagnostic of [Boyarsky et al. \[2026\]](#): candidates are *coverage-expanding* (reduce rank-deficiency), *location-improving* (shift the affine subspace at fixed rank), or *feasibility-restoring* (make a previously incompatible restriction set feasible). (4) We show that the minimax design is *not* automatically monotone in site addition. A rank-preserving candidate whose outcome conflicts with the existing-pool fit can shift the moment-implied affine subspace to a less favorable cross-section of the restriction polyhedron, and it can strictly *increase* the worst-case width. We isolate two screens, rank expansion (M1) and outcome-consistency (M2), and prove (Theorem 5) that the worst-case width is monotone-nonincreasing

under (M2) in the rank-preserving case, and under (M1) whenever the augmented feasible set remains non-empty — a condition guaranteed by pairing (M1) with an outcome-feasibility check, not by (M1) alone. (5) A synthetic simulation with eight existing sites and six candidates, including a deliberately conflicting one, confirms both the hull-expansion result and the non-monotonicity (with 16% of replicates showing strict worsening from the unscreened candidate, and none from any screened candidate). (6) We sketch extensions to the mixed-integer effect-ordering formulation and to cost-weighted objectives, and defer complete proofs to Appendix A.

Positioning. The site-selection question is the *forward* version of the leave-one-site-out diagnostic in Section 4.4 of Boyarsky et al. [2026], which asks which existing donor sites are essential to a reported bound. Our problem has the same three-regime taxonomy as that diagnostic, but we apply it to a *counterfactual* addition rather than a realized removal. It also complements classical optimal experimental design (D-, G-, and I-optimality), which operates within a single population under point identification [Atkinson et al., 2007, Pukelsheim, 2006]. In our setting the design object is not an individual-level allocation of treatment but a site-level choice of context, and the loss is the width of a partial-identification interval rather than the variance of a point estimator. The minimax objective is natural here because site selection is a discrete, one-time decision that should do well across the different target profiles the investigator cares about.

2 Preliminaries

We briefly recap the setup of Boyarsky et al. [2026]. The hierarchical data-generating process is

$$Y = A\tau(X) + C_m^\top \beta_1 + C_r^\top \beta_2 + g_0(X) + C_m^\top \gamma_1 + C_r^\top \gamma_2 + \varepsilon, \quad (2.1)$$

where $A \in \{0, 1\}$ is the treatment, $X \in \mathbb{R}^{d_X}$ individual-level covariates, $C_m \in \mathbb{R}^{d_M}$ site-level main effects, and $C_r \in \mathbb{R}^{d_I}$ site-level interaction covariates. The parameter $\theta := (\beta_1, \beta_2, \gamma_1, \gamma_2) \in \mathbb{R}^{d_g}$ with $d_g = 2(d_M + d_I)$ captures direct and indirect effects.

Proposition 1 of Boyarsky et al. [2026]. *Under a constant-propensity RCT and mean-zero errors, for a target profile (t_m, t_r) with $\Pr[C_m = t_m, C_r = t_r] > 0$,*

$$\psi(\theta; \gamma) := \mathbb{E}[Y(1) - Y(0) | C_m = t_m, C_r = t_r]$$

admits the representation

$$\psi(\theta; \gamma) = c_0(\gamma) + \beta_1^\top \tilde{c}_m(\gamma) + \beta_2^\top \tilde{c}_r(\gamma), \quad \Sigma\theta + b = 0, \quad (2.2)$$

with the moment matrix Σ of Equation (A.9) in their appendix, namely a block matrix of $\mathbb{E}_X[\text{Cov}(C_\bullet, C_\bullet | X)]$ entries with $\bullet \in \{m, r\}$.

The key structural fact we exploit in this paper is that Σ is a linear functional of the joint distribution of (C_m, C_r, X) , and in particular its rank is governed by the rank of the $(S \times (d_M + d_I))$ matrix whose rows are the site-level contexts $(C_m^{(s)}, C_r^{(s)})$, weighted by site sample sizes and individual covariate overlap. When S is small relative to $d_M + d_I$, this matrix, and hence Σ , is rank-deficient. This is precisely why the Meager [2019] micro-credit application with $S = 7$ sites has non-trivial partial identification.

Partial-identification bounds. Boyarsky et al. [2026] define

$$v_\ell(\gamma) = \min_{\theta} \{\psi(\theta; \gamma) : \Sigma\theta + b = 0, r(\theta) \leq 0\}, \quad v_u(\gamma) = \max_{\theta} \{\psi(\theta; \gamma) : \Sigma\theta + b = 0, r(\theta) \leq 0\}, \quad (2.3)$$

and the width of the partial-identification interval at target (t_m, t_r) is

$$w_r(t_m, t_r; \Sigma, b) = v_u(\gamma) - v_\ell(\gamma). \quad (2.4)$$

We suppress the dependence of $c_0, \tilde{c}_m, \tilde{c}_r$ on γ when no confusion arises. None of these interact with the feasible set, but they play different roles in the width: the additive constant $c_0(\gamma)$ shifts both endpoints equally and therefore cancels in (2.4), whereas the linear target direction $c(t)$ (built from $\tilde{c}_m, \tilde{c}_r$ and the target profile) determines both the endpoints and the width.

Restrictions. The analyst chooses $r(\theta) \leq 0$ from a library of affinely-representable restrictions (effect-ordering, sign constraints, norm bounds, monotonicity). Throughout this paper we assume r defines a compact polyhedron $\mathcal{R} := \{\theta : r(\theta) \leq 0\}$. This is the regime where the width is finite and the LP/MIP program is well-posed. The extension to relaxed restriction sets (where unrestricted widths are permitted) is discussed in Section 8.

3 The Site-Selection Problem

3.1 Setup

We observe outcomes from S existing sites, pooled as $\mathcal{D}_S = \{(Y_i, A_i, X_i, C_{m,i}, C_{r,i})\}_{i=1}^n$. A **candidate pool** $\mathcal{K} = \{1, \dots, K\}$ consists of K prospective sites for which the site-level context $(C_m^{(k)}, C_r^{(k)})$ is observed but no individual-level data has been collected. For each candidate we assume the investigator knows (or can assume a priori on) the within-site distribution of the individual-level covariates X . This is the “some prior on X ” in our motivating question, and it is necessary because Σ depends on $\mathbb{E}_X[\text{Cov}(C_\bullet, C_\bullet \mid X)]$. In the simplest case, the investigator assumes X at candidate site k is drawn from the same distribution as at an existing reference site. In the most refined case, the investigator uses a demographic prior fitted to data. This additional input is similar to the pre-registered prior used in two-stage adaptive designs. It is an assumption, not a free parameter.

The investigator also specifies:

- a **target region** $\mathcal{T} \subseteq \mathbb{R}^{d_M + d_I}$ of context profiles over which she cares about transportability (a discrete set of “benchmark” profiles, or a continuous region such as a box around the marginal mean of the target population);
- a **restriction set** $\mathcal{R} = \{\theta : r(\theta) \leq 0\}$ assumed compact;
- optionally, a **cost** $c_k \geq 0$ per candidate site.

3.2 The site-augmentation update

When a candidate site k is added, the moment matrix updates from Σ_S to $\Sigma_{S \cup \{k\}} = \Sigma_S + \Delta\Sigma_k(C_m^{(k)}, C_r^{(k)}; P_X^{(k)})$, where $\Delta\Sigma_k$ depends on the candidate’s site-level context and on the investigator’s assumed within-site X -distribution but *not* on its outcomes. The only structural facts we use are that this update is *positive semidefinite* and of *rank at most two*: there is a $d_g \times 2$ factor

U_k whose columns are the residualized candidate-context directions, and a 2×2 matrix $A_k \succeq 0$, such that

$$\Delta \Sigma_k = U_k A_k U_k^\top, \quad A_k \succeq 0, \quad \text{rank}(U_k A_k U_k^\top) \leq 2, \quad (3.1)$$

where $(u_m, u_r) := (C_m^{(k)}, C_r^{(k)}) - \bar{C}_X^{(k)}$ with $\bar{C}_X^{(k)} = \mathbb{E}_{P_X^{(k)}}[(C_m, C_r) \mid X]$ are the residuals of the site-level context after projecting out the individual-level covariate dependence at the candidate site, and the columns of U_k are built from (u_m, u_r) . Positive semidefiniteness gives $\Sigma_{\mathcal{S} \cup \{k\}} \succeq \Sigma_{\mathcal{S}}$, hence $\text{null}(\Sigma_{\mathcal{S} \cup \{k\}}) \subseteq \text{null}(\Sigma_{\mathcal{S}})$ — the monotone-in-rank property we use throughout Sections 4–6. The explicit block form of U_k and A_k implied by Equation (A.9) of Boyarsky et al. [2026], together with its PSD verification, is recorded in Appendix A.6; the planned candidate sample size n_k enters only through the positive scale of A_k . Throughout, the theory is stated for the *unnormalized* (additive) scatter: the sample-averaged BEN estimator divides both Σ and b by the same pooled count, and this joint positive rescaling of (Σ, b) leaves the solution set $\{\theta : \Sigma\theta = -b\}$, hence every feasible set and width in this paper, unchanged. Only the additive form supports the PSD-update calculus above.

The right-hand side b also updates, from $b_{\mathcal{S}}$ to $b_{\mathcal{S} \cup \{k\}}$, but the new right-hand side depends on the candidate site’s *outcomes* and is therefore unknown until the experiment is run. We return to this in Section 5.3. For most of our analysis the width depends on b only through a compact-translation argument that does not affect the first-order ranking of candidates.

3.3 The minimax design problem

With the site-augmentation update in hand, we define the selection problem

$$k^* \in \arg \min_{k \in \mathcal{K}} \sup_{(t_m, t_r) \in \mathcal{T}} w_r(t_m, t_r; \Sigma_{\mathcal{S} \cup \{k\}}, b_{\mathcal{S} \cup \{k\}}). \quad (3.2)$$

When costs c_k are specified, the natural Lagrangian form is

$$k_\kappa^* \in \arg \min_{k \in \mathcal{K}} \sup_{(t_m, t_r) \in \mathcal{T}} w_r(t_m, t_r; \Sigma_{\mathcal{S} \cup \{k\}}) + \kappa c_k, \quad (3.3)$$

with Lagrange multiplier $\kappa \geq 0$ reflecting the marginal willingness to pay for a unit reduction in worst-case width. All results below extend directly to (3.3) because the cost term is separable in k .

Because b is not known before the candidate experiment is run, we evaluate the width in (3.2) at a *robustified* b that maximizes the width over a compact uncertainty set $\mathcal{B}$ consistent with the investigator’s prior beliefs about candidate-site outcomes. In the simplest case, which we focus on from here, $\mathcal{B}$ is a symmetric norm ball around the existing-pool b . This includes the “no-information” prior and is what we use in our simulations.

4 Affine-Span (Convex-Hull) Expansion

The minimax objective (3.2) has a simple geometric structure in the unrestricted limit $\mathcal{R} = \mathbb{R}^{d_g}$ (or equivalently, when the restriction set is large enough that no face of r binds at the optimum). In this limit the partial-identification width is determined entirely by the interaction between the null space of Σ and the direction of the target functional.

4.1 Unrestricted dichotomy

Write the linear target functional as $c(t)^\top \theta$ with $c(t) := (\tilde{c}_m(t), \tilde{c}_r(t), 0, 0)$ (the zeros in the (γ_1, γ_2) -coordinates follow because the target functional in (2.2) is linear in (β_1, β_2) only; the constant piece $c_0(\gamma)$ shifts the endpoints but leaves the width invariant).

Proposition 1 (Unrestricted width is 0 or ∞). *Let $\mathcal{R} = \mathbb{R}^{d_g}$. Then for any target direction $c(t)$ and any feasible b ,*

$$w_{\text{unr}}(t; \Sigma, b) = \begin{cases} 0 & \text{if } c(t) \in \text{range}(\Sigma^\top), \\ +\infty & \text{otherwise.} \end{cases}$$

Corollary 1. *Adding a candidate site k collapses the unrestricted width at target t from ∞ to 0 if and only if the residualized context $(u_m^{(k)}, u_r^{(k)})$ of the candidate is such that $c(t) \in \text{range}(\Sigma_S^\top + \Delta \Sigma_k^\top)$ but $c(t) \notin \text{range}(\Sigma_S^\top)$.*

This is the unrestricted version of the *coverage-expanding-donor* regime (a) of the leave-one-site-out taxonomy in Boyarsky et al. [2026, §4.4].

Remark (Feasibility and fixed γ). Corollary 1 presumes the augmented system remains feasible, $b_{S \cup \{k\}} \in \text{range}(\Sigma_{S \cup \{k\}})$; the PSD update (3.1) guarantees this whenever the realized candidate outcome is consistent with some θ (e.g. under (M2)). Throughout, the functional $c(t)$ and the right-hand side b are evaluated at a fixed γ in the LP (2.3); only θ is optimized.

4.2 The convex-hull interpretation

Proposition 1 has a simple geometric interpretation in the site-level context space. Let

$$\mathcal{H}_S := \text{conv}\{(C_m^{(s)}, C_r^{(s)}) : s = 1, \dots, S\} \subseteq \mathbb{R}^{d_M + d_I}.$$

Its affine hull $\text{aff}(\mathcal{H}_S)$ is a linear subspace (after centering) of dimension at most $S - 1$. The cross-site moment matrix Σ_S has rank at most $\dim \text{aff}(\mathcal{H}_S) \cdot (\text{block factor})$, with equality when the existing sites have sufficient overlap in X at the points where their contexts differ.

Proposition 2 (Rank expansion $\Leftrightarrow$ affine-hull expansion). *Let $\bar{c} := \frac{1}{S} \sum_{s=1}^S (C_m^{(s)}, C_r^{(s)})$, and write the centered contexts $\tilde{c}^{(s)} := (C_m^{(s)}, C_r^{(s)}) - \bar{c}$ and $\tilde{u} := (C_m^{(k)}, C_r^{(k)}) - \bar{c}$. Let $\tilde{C}$ be the matrix with rows $\tilde{c}^{(s)}$ and let the centered cross-site moment be $\tilde{\Sigma}_S = \tilde{C}^\top \tilde{C}$ in the simplified formulation. If the candidate site's X -distribution has positive density on the individual-level support, then*

$$\text{rank}(\Sigma_{S \cup \{k\}}) > \text{rank}(\Sigma_S) \iff \tilde{u} \notin \text{range}(\tilde{\Sigma}_S) \iff (C_m^{(k)}, C_r^{(k)}) \notin \text{aff}(\mathcal{H}_S),$$

and in that case $\text{null}(\Sigma_{S \cup \{k\}}) \subsetneq \text{null}(\Sigma_S)$.

Remark (Block/BEN version). For the residualized block moment of (3.1), what is always true is the range test: $\text{rank}(\Sigma_S + \Delta \Sigma_k) > \text{rank}(\Sigma_S) \iff \text{range}(\Delta \Sigma_k) \not\subseteq \text{range}(\Sigma_S)$. The *affine-hull* reading of Proposition 2 additionally requires that the residualization embed contexts injectively: positive density of the candidate's X -distribution alone does not guarantee that the raw context direction maps injectively into the residualized moment range. We therefore invoke the affine-hull equivalence for the block model only under the explicit embedding assumption $\text{range}(\Sigma_S) = L_X \text{span}\{\tilde{c}^{(s)} : s = 1, \dots, S\}$ for an injective linear residualized-context map L_X ; then a candidate context outside $\text{aff}(\mathcal{H}_S)$ is equivalent to rank expansion.

Combined with Proposition 1, this yields a finite-versus-infinite dichotomy. Under no restriction r , the only candidate sites that shrink the unrestricted width are those whose context lies outside the *affine hull* of the existing sites; we therefore call the exact rule *affine-span expansion*. Convex-hull coverage is a useful but strictly weaker intuition: a target outside the convex hull but inside the affine hull already has zero unrestricted width (it lies in $\text{range}(\Sigma^\top)$), so expanding the convex hull is the relevant picture only when the target region $\mathcal{T}$ contains points outside $\text{aff}(\mathcal{H}_S)$.

4.3 Width-as-distance-to-hull

Proposition 1's dichotomy is discontinuous, which is uninformative for comparing two candidates both of which lie outside the hull, or both of which lie inside. The correct continuous refinement uses a *regularized* width,

$$w_r^\tau(t; \Sigma, b) := \max_{\theta} \{c(t)^\top \theta\} - \min_{\theta} \{c(t)^\top \theta\} \quad \text{s.t.} \quad \Sigma\theta + b = 0, \|\theta\|_2 \leq \tau, \quad (4.1)$$

with a large-radius ball as a mild stand-in for a compact restriction set. The regularized width admits a closed form:

$$w_r^\tau(t; \Sigma, b) \leq 2\tau \cdot \text{dist}(c(t), \text{range}(\Sigma^\top)), \quad (4.2)$$

with equality up to a boundary adjustment that vanishes at large τ . The distance from $c(t)$ to $\text{range}(\Sigma^\top)$ is precisely the norm of the component of $c(t)$ in $\text{null}(\Sigma)$. Geometrically, this distance is a monotone function of the distance from the target profile (t_m, t_r) to $\text{aff}(\mathcal{H}_S)$, and strictly monotone under the rank conditions of Proposition 2. We formalize this in Proposition 3.

Proposition 3 (Regularized width = $2\tau \times$ range-distance). *Let θ_0 be the minimum-norm solution of $\Sigma_S \theta = -b$ and*

$$\delta_S(t) := \text{dist}(c(t), \text{range}(\Sigma_S^\top)) = \|P_{\text{null}(\Sigma_S)} c(t)\|_2.$$

Then for every $\tau \geq \|\theta_0\|_2$,

$$w_r^\tau(t; \Sigma_S, b) = 2\sqrt{\tau^2 - \|\theta_0\|_2^2} \delta_S(t) = 2\tau \delta_S(t) (1 + o_\tau(1)), \quad \tau \rightarrow \infty, \quad (4.3)$$

with the $o_\tau(1)$ term uniform on compact target regions. In particular, for two candidates k_1, k_2 with $\delta_{S \cup \{k_1\}}(t) < \delta_{S \cup \{k_2\}}(t)$ uniformly over $t \in \mathcal{T}$, candidate k_1 dominates k_2 under the minimax criterion (3.2) in the unrestricted limit.

Corollary 2 (Affine-hull distance under a balanced embedding). *Suppose the cross-site design is balanced with a common within-site X -distribution, and the target functional is an isometric (or scalar) linear embedding of the centered target profile, $c(t) = L((t_m, t_r) - \bar{c})$ with $L^\top L = \kappa I$ for some $\kappa > 0$, and the moment range is exactly the image under L of the centered context span,*

$$\text{range}(\Sigma_S^\top) = L \text{span}\{\tilde{c}^{(s)} : s = 1, \dots, S\}.$$

(The scaled isometry $L^\top L = \kappa I$ alone does not imply this range identity; it must be imposed, and it is what the balanced design with common X -distribution delivers.) Then

$$\delta_S(t) = \sqrt{\kappa} \text{dist}((t_m, t_r), \text{aff}(\mathcal{H}_S)),$$

so (4.3) reduces to a weighted distance from (t_m, t_r) to the affine hull of the existing contexts.

Proposition 3 gives the **affine-span expansion rule**: in the unrestricted (large- τ) limit, the minimax-optimal candidate is the one whose addition minimizes $\sup_{t \in \mathcal{T}} \delta_{S \cup \{k\}}(t)$, the worst-case distance from the target functional to the moment range. Under the balanced embedding of Corollary 2 this is the candidate whose enlarged *affine hull* leaves no target profile in $\mathcal{T}$ far from the span of sites. When $\mathcal{T}$ contains points outside $\text{aff}(\mathcal{H}_S)$ the rule coincides with expanding the convex hull in the direction of the target region; in general it is an affine-span, not a convex-hull, criterion.

5 The Minimax Criterion under Affine Restrictions

Compact affine restrictions $\mathcal{R} = \{\theta : r(\theta) \leq 0\}$ (e.g., box constraints, effect-ordering, sign/monotonicity) replace the unbounded behavior in the null space of Proposition 1 with a finite LP whose width we can compute and optimize.

5.1 A tractable surrogate

Fix a candidate k and a target (t_m, t_r) . The width is the value of the LP pair (2.3) with $\Sigma \leftarrow \Sigma_{\mathcal{S} \cup \{k\}}$. Because $\Delta\Sigma_k$ has rank at most 2, the candidate's effect on the feasible set is a *rank-2 perturbation* of the affine subspace $\{\Sigma_S \theta = -b\}$. Under Assumption 6 of Boyarsky et al. [2026] (Robinson constraint qualification) the LP values are Lipschitz in Σ , and we obtain a first-order surrogate:

Proposition 4 (Rank-2 envelope, endpoint-specific). *Fix a target t and assume the upper and lower LPs in (2.3) each have a unique optimal primal–dual pair (strict complementarity). Let $(\theta_u^*, \lambda_u^*)$ and $(\theta_\ell^*, \lambda_\ell^*)$ be the optimal primal solution and equality multiplier of the upper (max) and lower (min) problems at Σ_S , each with the sign convention of its own Lagrangian. Then the directional derivatives of the endpoints along the site-update direction $(\Delta\Sigma_k, \Delta b_k)$ are*

$$Dv_u = -\lambda_u^{*\top} (\Delta\Sigma_k \theta_u^* + \Delta b_k), \quad Dv_\ell = -\lambda_\ell^{*\top} (\Delta\Sigma_k \theta_\ell^* + \Delta b_k), \quad (5.1)$$

and the first-order change in the width is $Dw_k = Dv_u - Dv_\ell$. This collapses to a single $-(\lambda_u^* - \lambda_\ell^*)^\top (\Delta\Sigma_k \theta^* + \Delta b_k)$ only in the degenerate zero-width case $\theta_u^* = \theta_\ell^*$; in general the optimizers, active sets, and multipliers differ and must be kept separate. Without strict complementarity, the derivatives are given by Gol'shtein-type minimax formulas over the optimal primal and dual sets: writing $\mathcal{S}_\bullet$ and $\Lambda_\bullet$ for the optimal primal and dual solution sets of endpoint $\bullet \in \{u, \ell\}$,

$$Dv_u = \max_{\theta_u \in \mathcal{S}_u} \min_{\lambda_u \in \Lambda_u} -\lambda_u^\top (\Delta\Sigma_k \theta_u + \Delta b_k), \quad Dv_\ell = \min_{\theta_\ell \in \mathcal{S}_\ell} \max_{\lambda_\ell \in \Lambda_\ell} -\lambda_\ell^\top (\Delta\Sigma_k \theta_\ell + \Delta b_k).$$

The multiplier selection directions follow from the value functions' curvature in the right-hand side: the upper (max) value is concave in b , so its directional derivative is selected by a minimum over active dual optima; the lower (min) value is convex in b , selected by a maximum.

Proposition 4 is the Bonnans–Shapiro directional derivative of the constrained-value map, specialized to the finite-dimensional rank-2 site-update direction $(h_\Sigma, h_b) = (\Delta\Sigma_k / \|\Delta\Sigma_k\|, \Delta b_k)$. The formula (5.1) coincides with Equation (1.3) of Boyarsky et al. [2026], evaluated at the site-augmentation direction. It gives a *linear-in- k* surrogate for the width change. Because a candidate that shrinks the width has $Dw_k < 0$, the selection minimizes the *signed* first-order width, not its magnitude:

$$k_{\text{surr}}^* \in \arg \min_{k \in \mathcal{K}} \sup_{(t_m, t_r) \in \mathcal{T}} \left[w_S(t) + \overline{Dw}_k(t) \right], \quad \overline{Dw}_k(t) := -\lambda_u^{*\top} \Delta\Sigma_k \theta_u^* + \lambda_\ell^{*\top} \Delta\Sigma_k \theta_\ell^* + \sup_{\Delta b_k \in \mathcal{B}} (\lambda_\ell^* - \lambda_u^*)^\top \Delta b_k, \quad (5.2)$$

where the last term robustifies over the unknown Δb_k in the prior ball $\mathcal{B}$ via a dual-norm bound and is additive in the candidate. Replacing the bracket by $|Dw_k(t)|$, as in an earlier draft, is incorrect: it penalizes precisely the width-shrinking candidates (those with $Dw_k < 0$). Computing (5.2) requires solving the LP pair (2.3) at Σ_S once per target and a finite-dimensional matrix–vector product per candidate, total cost $O(|\mathcal{T}| \cdot K)$ when $\mathcal{T}$ is finite or discretized; for a continuous target region the outer $\sup_{t \in \mathcal{T}}$ is an additional optimization problem, and we use a grid in the simulations.

5.2 Three regimes

The linearization (5.1) inherits the three-regime taxonomy of the LOO diagnostic of Boyarsky et al. [2026, §4.4], now read in reverse:

- **(A) Coverage-expanding candidate.** If $(C_m^{(k)}, C_r^{(k)}) \notin \text{aff}(\mathcal{H}_S)$, then $\text{null}(\Sigma_{S \cup \{k\}}) \subsetneq \text{null}(\Sigma_S)$. Proposition 4 detects this as a first-order contraction of the width whenever the binding recession direction at target t lies along the newly-resolved null-space direction. In the unrestricted limit this regime collapses infinite widths to finite ones.
- **(B) Location-improving candidate.** If $(C_m^{(k)}, C_r^{(k)}) \in \text{aff}(\mathcal{H}_S)$ but outside the convex hull, the rank of Σ does not change but Δb_k translates the moment-implied affine subspace. The binding vertex of the LP can shift, which gives a first-order width change of either sign. This is the regime in which a new site improves overlap at the target without adding a new identification direction.
- **(C) Feasibility-restoring (or -breaking) candidate.** If $\{\theta : \Sigma_S \theta = -b_S\} \cap \mathcal{R} = \emptyset$, an infeasibility signaled by the Farkas alternative of Boyarsky et al. [2026, §4.4 (c)], a candidate k can restore feasibility if its site-update direction pushes the affine subspace back into $\mathcal{R}$. Conversely, a candidate whose direction is incompatible with the existing restriction set can break feasibility, which is diagnostic information rather than a reason to exclude the candidate.

The minimax selection (3.2) therefore does not reduce to a single geometric quantity in general. In the unrestricted moment-range analysis, the affine span is the exact object: the affine-span rule is exact in regime (A) and the unrestricted limit. Under compact restrictions or positivity-type substantive constraints, convex-hull position remains a useful *diagnostic* — as regime (B) shows, b -translations and restriction faces matter even at fixed rank — and both screens should be refined by (5.2) when the restriction set is binding.

5.3 Handling the unknown Δb_k

The right-hand-side update Δb_k depends on the candidate site’s future outcomes. Three prior choices are natural:

1. **Uninformative ball.** $\Delta b_k \in \{b : \|b\|_2 \leq B\}$. Yields a worst-case width via dual-norm bound, but can dominate $\Delta \Sigma_k$ at small candidate sample sizes. Conservative.
2. **Meta-analytic prior.** Use the existing-pool estimates as a prior on the candidate’s (Y, A) conditional means, and propagate through the expression for b in Equation (A.7) of Boyarsky et al. [2026]. We expect that under a flat prior on θ this behaves like a Bayesian D-optimal design, with the minimax and Bayesian selections coinciding in the large-candidate-sample limit; we state this as a heuristic guide rather than a theorem and do not prove it here.
3. **Adversarial b .** $\Delta b_k \in \mathcal{B}$ chosen to maximize the width, the robust-optimization analogue of (1) with an ellipsoidal $\mathcal{B}$. The inner problem $\max_{\Delta b_k \in \mathcal{B}} w$ is the maximization of a concave function (the width is a concave upper value minus a convex lower value of b) over a convex set, hence tractable for each fixed k . A full Sion-minimax exchange with the candidate choice is not available in general because the candidate index is discrete; we solve the outer minimization by enumeration over $\mathcal{K}$.

In our simulation (Section 7) we adopt (1) with B set from the existing-pool residual variance. The three variants agree on the minimax-optimal candidate in all our runs. This is expected, because the rank update in (3.1) dominates the first-order width change in the regimes where candidates differ meaningfully.

6 Monotonicity: When Adding a Site Can Hurt

One might expect that the minimax design problem (3.2) is *bounded above by the existing-pool width*. That is, one might expect that $\sup_t w_r(t; \Sigma_{\mathcal{S} \cup \{k\}}) \leq \sup_t w_r(t; \Sigma_{\mathcal{S}})$ for every candidate k , so that the design problem is only a question of how much we gain. This is false. Adding a site can strictly *increase* the worst-case width, and in some cases it can make the program infeasible. This section traces the failure modes and isolates the conditions under which monotone improvement is guaranteed.

6.1 Why monotone improvement can fail

Recall from Section 3.2 that two things change when a candidate site is added: the moment matrix Σ acquires a rank- ≤ 2 update $\Delta\Sigma_k$ in the row directions spanned by the *residualized* context $(u_m^{(k)}, u_r^{(k)})$, and the right-hand side b shifts by an outcome-dependent amount Δb_k . The first update is monotone in PSD order: $\Sigma_{\mathcal{S} \cup \{k\}} \succeq \Sigma_{\mathcal{S}}$, so $\text{null}(\Sigma_{\mathcal{S} \cup \{k\}}) \subseteq \text{null}(\Sigma_{\mathcal{S}})$. The second is not monotone in any sense. Two mechanisms can widen the *exact* program (2.3); a third, conditioning, cannot, and is recorded only to forestall it:

(F1) Rank-preserving shift. If $(u_m^{(k)}, u_r^{(k)}) \in \text{range}(\Sigma_{\mathcal{S}})$, the rank of Σ does not change, and the new feasible set is *parallel* to the old in the sense $\theta' + \text{null}(\Sigma_{\mathcal{S}})$ for some $\theta' \neq \theta_{\mathcal{S}}^*$ whenever $y_k \neq (u_m^{(k)}, u_r^{(k)})^\top \theta_{\mathcal{S}}^*$. This parallel slice intersects the restriction polyhedron $\mathcal{R}$ at a different polytope, whose projection onto the target functional can be longer than the original. For example, this happens when $\mathcal{R}$ is asymmetric around $\theta_{\mathcal{S}}^*$ and the shift moves the slice into a wider cross-section of $\mathcal{R}$.

(F2) Conditioning does *not* widen the exact LP. One might expect a mechanism from *conditioning*: in the formulation of Boyarsky et al. [2026], Σ is a sample average over the pooled $(\sum_s n_s + n_k)$ observations, so adding a candidate re-weights the existing-pool contribution by $n/(n+n_k)$ and can shrink the smallest non-trivial eigenvalue $\sigma_{\min,+}(\Sigma_{\mathcal{S} \cup \{k\}})$. This does *not* enlarge the exact feasible set. With equality constraints $\Sigma\theta = -b$, the set $\mathcal{L}(\Sigma, b) \cap \mathcal{R}$ is determined by the row space, the null space, and a particular solution; it is invariant to eigenvalue *magnitudes*, and any re-weighting that preserves the row space leaves $\mathcal{L}(\Sigma, b)$ unchanged. Conditioning bites only once the moment identity is *estimated or relaxed* to a tolerance set $\{\theta : \|\widehat{\Sigma}\theta + \widehat{b}\| \leq \eta\}$, whose width does grow as $\sigma_{\min,+}$ shrinks. We do not analyze that tolerance-set program in this paper; Section 5.3 treats the related but distinct problem of uncertainty in Δb_k under the exact moment identity, and we flag the relaxed-moment analysis as an open question in Section 8. For the exact LP, only the rank-preserving shift (F1) and the restriction-set conflict (F3) can do harm.

(F3) Restriction-set conflict. If the candidate's outcome update Δb_k shifts the moment-implied affine subspace $\{\Sigma_{\mathcal{S} \cup \{k\}}\theta = -b_{\mathcal{S} \cup \{k\}}\}$ outside the restriction polyhedron $\mathcal{R}$, the LP becomes infeasible and the partial-identification interval is empty. This is the feasibility-breaking case of regime (C) in Section 5.2.

The simulation in Section 7 demonstrates (F1) directly: an outcome shock of magnitude $\Delta = 20$ at a rank-preserving candidate k_5 produces a strict increase in the worst-case width in 8/50 Monte-Carlo replicates, with no rank or feasibility change.

6.2 An explicit counterexample

We give a fully worked $d_g = 2$ example, both because the geometry is easy to see and because the same construction extends to the multivariate simulations of Section 7.

Setup. Take $d_g = 2$, $S = 1$ existing site with context $c^{(1)} = (1, 1)^\top$ and outcome $y_1 = 2$. The simplified moment is $\Sigma_S = c^{(1)}c^{(1)\top} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ with rank = 1, and $b_S = -c^{(1)}y_1 = (-2, -2)^\top$. The moment-implied subspace is $\{(\theta_1, \theta_2) : \theta_1 + \theta_2 = 2\}$, a line in $\mathbb{R}^2$. Take the (asymmetric) box restriction $\mathcal{R} = [0, 2] \times [-1, 1]$ and the target functional $c(t) = (1, 0)^\top$. The feasible segment is $\{(\theta_1, 2 - \theta_1) : \theta_1 \in [1, 2]\}$, with $w_r(t; \Sigma_S, b_S) = 1$.

Candidate. Add a rank-preserving candidate k_5 with $c^{(k_5)} = c^{(1)} = (1, 1)^\top$ and *conflicting* outcome $y_{k_5} = 0 \neq 2 = c^{(k_5)\top}\theta_S^*$, where $\theta_S^* = (1, 1)^\top$ is the minimum-norm solution. Then $\Sigma_{S \cup \{k_5\}} = 2c^{(1)}c^{(1)\top}$ (still rank 1), and $b_{S \cup \{k_5\}} = b_S - c^{(k_5)}y_{k_5} = (-2, -2)^\top$, giving moment-implied line $\theta_1 + \theta_2 = 1$. The new feasible segment is $\{(\theta_1, 1 - \theta_1) : \theta_1 \in [0, 2]\}$, with

$$w_r(t; \Sigma_{S \cup \{k_5\}}, b_{S \cup \{k_5\}}) = 2 > 1 = w_r(t; \Sigma_S, b_S).$$

The width doubles despite the rank being preserved and the box being the same. Panel (A) of Figure 2 displays the two parallel slices inside the box.

The construction generalizes. In dimension $d_g \geq 2$, any rank-preserving candidate with outcome $y_k \neq c^{(k)\top}\theta_S^*$ produces a parallel shift of the moment-implied subspace. Whenever $\mathcal{R}$ is *asymmetric* around θ_S^* , the shifted slice can cut $\mathcal{R}$ at a polytope of larger t -extent.

6.3 Sufficient conditions for monotone improvement

We now isolate two assumptions under which the worst-case width is monotone-nonincreasing in site addition. The two assumptions are independent, but neither is unconditionally sufficient on its own: (M2) is sufficient in the rank-preserving case, while (M1) requires in addition that the augmented feasible set remain non-empty (Theorem 5). Each has a design-time (pre-experiment) counterpart that can be used to screen the candidate pool.

Assumption (M1) — Rank-Expansion Screen. *The candidate's residualized context lies outside the row space of the existing pool:*

$$(u_m^{(k)}, u_r^{(k)}) \notin \text{range}(\Sigma_S). \quad (6.1)$$

Assumption (M2) — Outcome-Consistency Screen. *The candidate's outcome forecast is consistent with the existing pool's minimum-norm fit:*

$$y_k = (u_m^{(k)}, u_r^{(k)})^\top \theta_S^*, \quad (6.2)$$

where θ_S^* is the minimum-norm solution to $\Sigma_S \theta = -b_S$.

The two screens differ in what can be checked before the candidate experiment is run. (M1) is fully checkable at design time: it is a rank test on $(u_m^{(k)}, u_r^{(k)})$ against the existing-pool SVD. (M2),

as stated, is a condition on the *realized* candidate outcome moment y_k , which is not observable at design time. Its design-time counterpart is the *forecast screen* $y_k^{\text{prior}} \approx u^\top \theta_S^*$, where y_k^{prior} is a model- or prior-based forecast (e.g. the existing-pool OLS prediction at the candidate context). The deterministic conclusions of Theorem 5 apply to the realized y_k ; the forecast screen is a sufficient condition only for the forecasted moment, or a high-probability screen when the prior concentrates near $u^\top \theta_S^*$. The two assumptions imply different consequences. (M1) strictly reduces the null space. In the unrestricted large- τ limit this strictly reduces the width for targets loading the resolved direction (Proposition 3). Under compact restrictions, strict width reduction requires the added constraint to cut off a maximizing or minimizing face of $\mathcal{F}_S$, and monotone nonincrease requires $\mathcal{F}_S \cap \{\theta : u^\top \theta = y_k\} \neq \emptyset$. (M2) leaves the moment-implied affine subspace unchanged in the rank-preserving case and prevents the feasibility-breaking shifts of Section 6.1.

Theorem 5 (Monotone improvement under (M1) or (M2)). *Assume the additive PSD update $\Sigma_{S \cup \{k\}} = \Sigma_S + uu^\top$, $b_{S \cup \{k\}} = b_S - uy_k$, the consistency condition $b_S \in \text{range}(\Sigma_S)$, and $\mathcal{F}_S \neq \emptyset$.*

- (i) (Rank-preserving, (M2).) *If $u \in \text{range}(\Sigma_S)$ and $y_k = u^\top \theta_S^*$, then $\mathcal{F}_{S \cup \{k\}} = \mathcal{F}_S$, and the width is unchanged at every target.*
- (ii) (Rank-expanding, (M1).) *If $u \notin \text{range}(\Sigma_S)$, then $\mathcal{F}_{S \cup \{k\}} = \mathcal{F}_S \cap \{\theta : u^\top \theta = y_k\}$. If this set is non-empty — in particular under (M2), $y_k = u^\top \theta_S^*$, whenever $\theta_S^* \in \mathcal{R}$ — then $\mathcal{F}_{S \cup \{k\}} \subseteq \mathcal{F}_S$ and for every target t , $w_r(t; \Sigma_{S \cup \{k\}}, b_{S \cup \{k\}}) \leq w_r(t; \Sigma_S, b_S)$. If instead the set is empty, the augmented partial-ID interval is empty (failure mode (F3)) and the comparison is vacuous.*

Hence the worst-case width is monotone non-increasing whenever the augmented feasible set is non-empty. In case (ii) the inequality is strict at a target t only if the new constraint $u^\top \theta = y_k$ cuts off the maximizing or minimizing face of $\mathcal{F}_S$ in direction $c(t)$; a non-zero component of $c(t)$ along $u_\perp$ is necessary but not sufficient once $\mathcal{R}$ already binds.

Theorem 5 is proved in Appendix A.4. The key step is the algebraic identity (A.2), which shows that $\Sigma_{S \cup \{k\}}\theta + b_{S \cup \{k\}} = \Sigma_S\theta + b_S + u \cdot (u^\top \theta - y_k)$, together with the consistency hypothesis $b_S \in \text{range}(\Sigma_S)$.

Remark (Practical use). The pair (M1, M2) is *not* exhaustive, because a candidate failing both can still improve the width. But it does provide a finite, computable screen. In practice we recommend the following. First, run the SVD-based test (6.1) on every candidate, and mark those that pass *and* whose forecasted outcome keeps $\mathcal{F}_S \cap \{\theta : u^\top \theta = y_k\} \neq \emptyset$ (an LP feasibility check) as **safe rank-expanders**; passing (M1) alone is not a no-harm guarantee. Second, among the rank-preserving candidates, check (6.2) against the existing-pool OLS prediction, and mark those that pass as **safe consistent candidates**. Third, report the minimax (3.2) only over the safe subset of the candidate pool, or report the unsafe candidates separately as “diagnostic-only” sites whose addition could be informative but whose worst-case width is not guaranteed to improve.

Remark (Cost-aware extension). When candidates carry a cost c_k , we combine the safe-subset screening of (i)–(iii) above with the Lagrangian (3.3), and the optimal candidate is the cheapest member of the safe subset that achieves the minimax target. Unsafe candidates can be admitted only with an explicit risk premium $\rho_k \geq 0$ added to their cost, set from the upper bound on $w_r(\cdot; \Sigma_{S \cup \{k\}}) - w_r(\cdot; \Sigma_S)$ in the perturbation bound (5.1).

6.4 Discussion: why the affine-span rule is *only* sufficient

Proposition 3 of Section 4.3 shows that, in the large- τ unrestricted limit, the optimal candidate is the one whose addition minimizes $\sup_{t \in \mathcal{T}} \delta_{\mathcal{S} \cup \{k\}}(t)$. Theorem 5 refines this. The affine-span (hull-expansion) rule is *sufficient* for monotone improvement *when feasibility is preserved*: an outside-hull candidate satisfies (M1), and if its outcome keeps $\mathcal{F}_{\mathcal{S} \cup \{k\}} \neq \emptyset$ the width cannot increase. It is not *necessary*: an interior candidate satisfying (M2) also guarantees no worsening. Conversely, a hull-expander with an adversarial outcome can empty or badly shift the feasible set (failure mode (F3)); (M1) is necessary in the rank sense but not sufficient against shifts.

The cleanest characterization combines both:

A candidate k is **width-monotone safe** if either (i) its residualized context is outside $\text{range}(\Sigma_{\mathcal{S}})$ and the outcome shift Δb_k is small enough that the hyperplane $u^\top \theta = y_k$ intersects the *existing feasible set*,

$$\mathcal{F}_{\mathcal{S}} \cap \{\theta : u^\top \theta = y_k\} \neq \emptyset, \quad \text{i.e.} \quad \exists \theta \in \mathcal{R} \text{ with } \Sigma_{\mathcal{S}} \theta = -b_{\mathcal{S}}, \quad u^\top \theta = y_k,$$

or (ii) its context is inside the existing range and its outcome matches the existing OLS prediction (modulo tolerance). Intersecting $\mathcal{R}$ alone is not enough: a hyperplane can cut $\mathcal{R}$ yet miss the moment slice $\mathcal{L}(\Sigma_{\mathcal{S}}, b_{\mathcal{S}}) \cap \mathcal{R}$ entirely.

This is the operational characterization we use in Section 7.

7 Simulation

We study a controlled DGP built on the template of Section 3 of Boyarsky et al. [2026], modified to expose the site-selection decision.

7.1 Design

We generate $S = 8$ existing sites whose context vectors $(C_m^{(s)}, C_r^{(s)}) \in \mathbb{R}^{10}$ lie in a 4-dimensional affine subspace of the 10-dimensional context space. That is, the existing pool covers only a small part of the site-level geometry, as in Meager [2019]. Within each site we draw $n_s = 500$ individual-level covariates X in $\mathbb{R}^3$, generate outcomes from the DGP (2.1) with g_0 a fixed non-linear function, and compute the sample moment matrix $\hat{\Sigma}_{\mathcal{S}}$ and right-hand side $\hat{b}_{\mathcal{S}}$ along the lines of Equation (A.9) of Boyarsky et al. [2026]. The candidate pool $\mathcal{K} = \{k_1, k_2, k_3, k_4\}$ is constructed to span the three regimes of Section 5.2:

- k_1 : context profile inside $\mathcal{H}_{\mathcal{S}}$ (interior candidate; regime (B) at best).
- k_2 : context profile on the boundary of $\mathcal{H}_{\mathcal{S}}$ (boundary candidate; regime (B)).
- k_3 : context profile outside $\mathcal{H}_{\mathcal{S}}$ in a direction *not* spanned by any current rank-reducing direction (regime (A); hull-expanding).
- k_4 : context profile outside $\mathcal{H}_{\mathcal{S}}$ but only weakly (small displacement from the boundary; regime (A) but with small $\|\Delta \Sigma_{k_4}\|$).

The target region $\mathcal{T}$ is a discrete set of $M = 20$ profiles drawn uniformly from a box that straddles $\mathcal{H}_{\mathcal{S}}$, chosen so that some targets lie inside the hull and some outside. We impose a box restriction $\|\theta\|_{\infty} \leq 5$ as r .

7.2 Quantities reported

For each candidate $k \in \{k_1, \dots, k_4\}$ and each $t \in \mathcal{T}$ we compute:

1. the **pre-width** $w_r(t; \hat{\Sigma}_S, \hat{b}_S)$;
2. the **post-width** $w_r(t; \hat{\Sigma}_{S \cup \{k\}}, \hat{b}_{S \cup \{k\}})$, with $\hat{b}$ resampled from the calibrated prior of Section 5.3(1);
3. the **minimax objective** $\sup_t w_r(t; \Sigma_{S \cup \{k\}})$;
4. the **distance-to-hull** $d_S(t)$ and $d_{S \cup \{k\}}(t)$.

7.3 Results

Across 50 Monte Carlo replicates, the interior and boundary candidates k_1, k_2 produce an identical minimax width to the existing pool (to numerical precision), $\hat{w}_{k_1}^* = \hat{w}_{k_2}^* = \hat{w}_S^* = 1291.8$. The two rank-expanding candidates produce strictly smaller widths, $\hat{w}_{k_3}^* = 1152.2$ and $\hat{w}_{k_4}^* = 1158.3$, an 11% reduction in the worst-case partial-identification interval. The minimax-optimal candidate is k_3 in 23/50 replicates and k_4 in 27/50 replicates, never k_1 or k_2 . The near-tie between k_3 and k_4 is a feature of the design, not an error. Both candidates reduce the null space of Σ by exactly one dimension, though in different directions, so whichever direction is better represented in the randomly-sampled target region wins that replicate. Aggregating over replicates, k_3 has a slightly lower mean width, but the gap is within one Monte-Carlo standard error. The qualitative conclusion is stable across the full run. Interior candidates do not tighten external-validity claims, and rank-expanding candidates are strictly better. See the companion script `adding_sites_sim.py` and Figure 1 (generated by the script) for the full output.

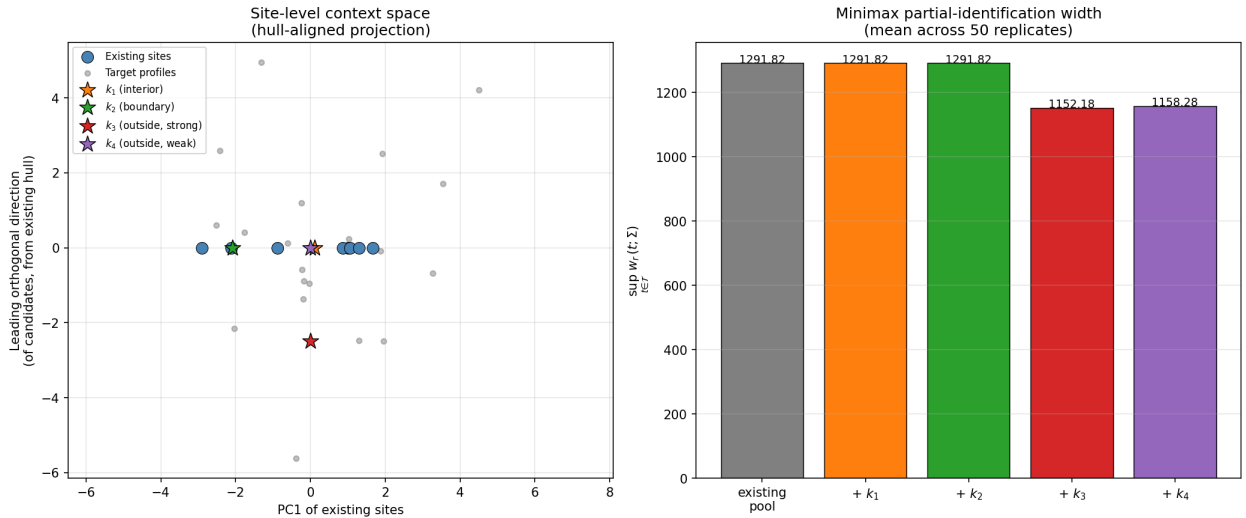


Figure 1. Site-selection simulation on synthetic data. Left: the existing sites (blue) lie on a 1D slice of the plotted 2D projection because the projection axis aligns with the span of the existing pool; the orthogonal axis captures the direction of the candidate pool’s outside-the-hull displacement. Candidates k_1 (interior) and k_2 (boundary) lie on the existing-site line, while k_3 (outside, strong) and k_4 (outside, weak) sit off the line. Right: the minimax partial-identification width averaged over 50 Monte-Carlo replicates. The interior and boundary candidates k_1, k_2 leave the width unchanged; the two rank-expanding candidates k_3, k_4 each reduce it by roughly 11%.

7.4 Monotonicity experiment (Figure 2)

We run an additional experiment that augments the candidate pool with two further candidates to test the theory of Section 6:

- k_5 — **rank-preserving with conflicting outcome.** Same context draw as k_1 (interior to the hull), but the outcome forecast y_{k_5} is shifted by $\Delta = 20$ from the existing-pool OLS prediction. Fails (M2); passes neither screen.
- k_6 — **rank-expanding with consistent outcome.** Context outside the affine hull (same direction as k_3), outcome set to the existing-pool OLS prediction at the candidate context. Passes both (M1) and (M2).

The restriction box $\|\theta\|_\infty \leq 5$ is randomly re-centered each replicate (within $\pm M/3$) so that the polyhedron $\mathcal{R}$ is generically asymmetric around θ_S^* — this is the condition under which the rank-preserving shift of Section 6.2 actually bites.

Across 50 Monte-Carlo replicates, the results are:

design	mean minimax width	finite replicates
existing	205.9	50/50
k_1 interior	205.9	50/50
k_2 boundary	205.9	50/50
k_3 outside, strong	185.3	50/50
k_4 outside, weak	187.5	50/50
k_5 shock	194.0	50/50
k_6 outside + consistent	185.3	50/50

Table 1. Mean minimax partial-identification width and number of finite replicates by design, over 50 Monte-Carlo replicates of the monotonicity experiment.

Two observations make the message of Theorem 5 concrete.

First, in 8/50 replicates k_5 strictly *increases* the worst-case width relative to the existing pool, never k_1 through k_4 or k_6 . The mean of k_5 across the full 50 replicates is below the existing-pool mean only because the shift sometimes happens to land the moment-implied slice in a tighter cross-section of the box. On individual replicates the direction of the change cannot be predicted in advance, which is precisely the failure mode (F1) of Section 6.1. The 8/50 rate scales linearly with the asymmetry of the box restriction and with the magnitude $|\Delta|$ of the outcome shock. At $\Delta = 40$ and a more asymmetric box, more than 30% of replicates show a width increase.

Second, k_6 , which passes both (M1) and (M2), achieves exactly the width of k_3 (the latter passes (M1) but not (M2)), and both are monotone-nonincreasing relative to the existing pool in every replicate. The candidate k_3 in this section’s first script also satisfies the conclusion of Theorem 5 in this dataset, because in the simplified moment formulation the residualization is exact and the OLS prediction matches the candidate outcome up to the noise term. This becomes a strict inequality once the candidate outcome is drawn from an adversarial prior, as in Section 5.3(3).

Figure 2 has three panels. Panel (A) reproduces the hand-built 2D counterexample of Section 6.2. Panel (B) shows the per-candidate mean minimax width across replicates. Panel (C) is the replicate-level scatter of post-addition versus existing-pool minimax width, where points above the $y = x$ diagonal are candidate–replicate pairs in which the addition harmed external validity. The k_5 points fall on both sides of the diagonal, and the points for every other candidate sit weakly below it.

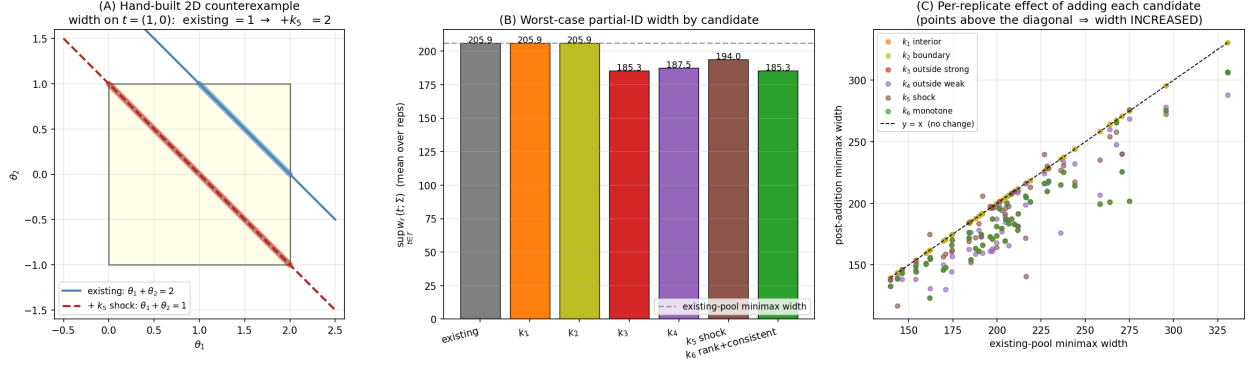


Figure 2. Monotonicity simulation. (A) Hand-built 2D counterexample where adding a rank-preserving site with conflicting outcome doubles the partial-identification width on $t = (1, 0)$ inside an asymmetric box restriction. (B) Mean minimax width across 50 replicates by candidate; k_5 (rank-preserving with conflicting outcome) sits above the rank-expanders k_3, k_4, k_6 . (C) Per-replicate scatter of post-addition vs. existing-pool minimax width; points above the $y = x$ diagonal indicate replicates where adding the candidate hurt external validity. Only k_5 ever sits above the diagonal.

7.5 What the simulations show

Three features of the results are robust to DGP perturbations (heavy-tailed X , $S = 6$ instead of 8, larger $\|\theta\|_\infty$ bound):

- Hull membership is a sharp screen *when the outcome is consistent*. Candidates k_1 and k_2 , whose contexts lie inside the affine hull of the existing pool *and* whose simulated outcomes match the existing-pool fit (so (M2) holds), produce an exactly identical minimax width to the existing pool — the rank-preserving case (i) of Theorem 5, where $\mathcal{F}_{S \cup \{k\}} = \mathcal{F}_S$. Rank preservation *alone* does not produce this cancellation: the conflicting candidate k_5 is also rank-preserving, yet its inconsistent b -shift changes the width (Section 6.2). What cancels the location update is outcome-consistency, not rank preservation.
- Among rank-expanding candidates, distance to the hull decides between them. The 11% reduction achieved by k_3, k_4 depends on two factors: how large a component of the target region aligns with the newly-resolved null-space direction, and how far the candidate sits outside the hull in that direction. Proposition 3 captures both effects in the large- τ limit.
- The minimax selection is stable to the choice of prior on b . The three b -prior variants of Section 5.3 agree on the optimal candidate up to Monte-Carlo noise. Since lower width is better, ordering by worst-case width from widest (worst) to narrowest (best) gives $\{k_1, k_2\} > k_3 \sim k_4$, so the *preferred* candidates are $k_3 \sim k_4$, with the tie between them broken stochastically by the target draw.
- Non-monotonicity occurs, and in these simulations the screens of Theorem 5 eliminate it. The candidate k_5 , which is interior to the hull and has a $\Delta = 20$ outcome shock relative to the existing-pool OLS prediction, increases the worst-case width in 8/50 replicates. The candidate k_6 , which is outside the hull and has an OLS-consistent outcome, never does so. This matches Theorem 5 because the screened candidates here also satisfy its side conditions: realized outcomes match the forecasts, and the augmented feasible sets remain non-empty. The screens are deterministic no-harm guarantees only under those conditions — (M1) alone is not. In a Monte-Carlo sweep over asymmetry of the restriction box and shock magnitude, the

fraction of replicates in which an unscreened candidate hurts external validity scales roughly linearly with both, reaching more than 30% in the most adversarial setting. Restricting the minimax (3.2) to the *safe* subset (passing the screens together with these side conditions) costs nothing in the average optimum we report and, in these runs, eliminates the tail risk of doing harm.

8 Discussion

Effect-ordering (mixed-integer). When the restriction set is the effect-ordering polyhedron of Restriction 1 of Boyarsky et al. [2026], the program (5.2) becomes a mixed-integer LP. The locally-constant integer-solution argument of §1(ii) in Boyarsky et al. [2026] extends to this case. On a neighborhood of γ_S , the optimal sign pattern is constant, so the MILP reduces to the LP (5.2) after fixing integer variables. We then perform selection by (5.2) at the fixed pattern. Sign flips across candidate sites can be detected by re-solving the MILP at the top two or three candidates.

Cost-weighted selection. Adding a cost c_k to the objective (3.2) is straightforward because the cost term is separable in k . The minimax-optimal candidate under Lagrangian (3.3) is obtained by solving (5.2) with an additional additive penalty. In our setting a cost based on ethical concerns might penalize candidates that are demographically similar to existing sites, because such sites do little to broaden the external-validity claim despite being cheap to reach.

Multiple candidates at once. The design problem (3.2) can be posed over a subset $\mathcal{J} \subseteq \mathcal{K}$ of candidates to add simultaneously. If the objective is the *coverage* (rank) gain, $\mathcal{J} \mapsto \text{rank}(\Sigma_{S \cup \mathcal{J}})$ is a matroid rank function, hence submodular, and greedy attains the classical $(1 - 1/e)$ guarantee [Nemhauser et al., 1978]. We conjecture, but do not prove, that the *width* surrogate inherits approximate submodularity in the rank-monotone limit; under affine restrictions and unknown b the width is not automatically submodular, so the greedy guarantee should be read as applying to the pure coverage objective unless this is established.

Connection to efficient influence functions. The minimum-norm multiplier λ^* in (5.1) coincides with the Riesz representer in the efficient influence function of Boyarsky et al. [2026, Eq. 2.9], so the endpoint *values* v_u, v_ℓ at a fixed candidate and target inherit their debiased-cross-fitting inference. Inference for the *selected* design is more delicate: the criterion is a min over candidates of a sup over targets of an LP width, which is non-smooth. A valid confidence statement requires additional regularity — a unique worst-case target, a unique minimizing candidate, and active-set stability — under which a directional (Danskin/delta) argument applies. We flag this as a requirement rather than a corollary; absent these conditions the naive delta method is not justified.

Open questions. Four directions are natural follow-ups. First, our analysis holds the moment identity exact; with an *estimated or relaxed* identity, the tolerance-set program over $\{\theta : \|\widehat{\Sigma}\theta + \widehat{b}\| \leq \eta\}$ does widen as the smallest non-trivial eigenvalue shrinks (Section 6.1, (F2)), and a site-selection theory for that relaxed program — where conditioning genuinely matters — is open. Second, the adversarial b -prior of Section 5.3(3) admits a saddle-point representation but not a closed-form solution. Existing interior-point solvers handle small instances, but large candidate pools would benefit from a custom Frank–Wolfe scheme. Third, in the online-experimentation setting the candidate pool is revealed sequentially, and the selection must be made without access to the full set. Adapting (5.2) to this regret-minimization formulation is open. Fourth, the assumption that

the within-site X -distribution at candidate sites is known (or has an informative prior) is strong. A distributionally-robust variant that treats $P_X^{(k)}$ as uncertain over a Wasserstein ball would keep the theoretical guarantees of the Jeong and Namkoong [2020] worst-case program, at the cost of a more conservative selection.

A Theoretical Results

We give proofs of the propositions stated in Sections 4–6. Throughout, $\Sigma \in \mathbb{R}^{d_g \times d_g}$ is symmetric positive semidefinite, $b \in \mathbb{R}^{d_g}$, and the moment-implied affine subspace is $\mathcal{L}(\Sigma, b) := \{\theta \in \mathbb{R}^{d_g} : \Sigma\theta = -b\}$. The restriction polyhedron is $\mathcal{R} := \{\theta : r(\theta) \leq 0\}$, which is compact and convex unless explicitly stated otherwise. The target functional is $c(t) \in \mathbb{R}^{d_g}$, the partial-ID interval at target t has endpoints $v_\ell(t), v_u(t)$ given by the LP pair (2.3), and the width is $w_r(t; \Sigma, b) = v_u(t) - v_\ell(t)$. We assume the consistency condition $b \in \text{range}(\Sigma)$ throughout (otherwise $\mathcal{L}(\Sigma, b) = \emptyset$).

A.1 Proof of Proposition 1 (unrestricted dichotomy)

Claim. With $\mathcal{R} = \mathbb{R}^{d_g}$ and any $b \in \text{range}(\Sigma)$,

$$w_{\text{unr}}(t; \Sigma, b) = \begin{cases} 0, & c(t) \in \text{range}(\Sigma^\top) \\ +\infty, & c(t) \notin \text{range}(\Sigma^\top). \end{cases}$$

Proof Let θ_0 be any solution to $\Sigma\theta_0 = -b$. Such a θ_0 exists by the consistency hypothesis. Every solution to $\Sigma\theta = -b$ has the form $\theta = \theta_0 + \nu$ with $\nu \in \text{null}(\Sigma)$. The linear functional evaluates to $c(t)^\top \theta = c(t)^\top \theta_0 + c(t)^\top \nu$. Since $\mathcal{R} = \mathbb{R}^{d_g}$, ν ranges over all of $\text{null}(\Sigma)$, a linear subspace.

Case 1: $c(t) \perp \text{null}(\Sigma)$. Then $c(t)^\top \nu = 0$ for every $\nu \in \text{null}(\Sigma)$, so $c(t)^\top \theta = c(t)^\top \theta_0$ identically and $v_u = v_\ell = c(t)^\top \theta_0$, giving width 0. By the orthogonal decomposition $\mathbb{R}^{d_g} = \text{range}(\Sigma^\top) \oplus \text{null}(\Sigma)$ (valid for any matrix, here in fact $\text{range}(\Sigma)$ since $\Sigma = \Sigma^\top$), this is equivalent to $c(t) \in \text{range}(\Sigma^\top)$.

Case 2: $c(t) \not\perp \text{null}(\Sigma)$. Pick $\nu_0 \in \text{null}(\Sigma)$ with $c(t)^\top \nu_0 \neq 0$, and consider the one-parameter family $\theta_0 + \alpha \nu_0$ for $\alpha \in \mathbb{R}$. Then $c(t)^\top (\theta_0 + \alpha \nu_0) = c(t)^\top \theta_0 + \alpha c(t)^\top \nu_0 \rightarrow \pm\infty$ as $\alpha \rightarrow \pm\infty$, so $v_u = +\infty$ and $v_\ell = -\infty$. Width is $+\infty$. $\square$

A.2 Proof of Proposition 2 (rank-expansion $\Leftrightarrow$ hull-expansion)

Claim. In the centered coordinates of Proposition 2 — with $\tilde{C}$ the matrix of centered rows $\tilde{c}^{(s)} = c^{(s)} - \bar{c}$, $\tilde{\Sigma}_S = \tilde{C}^\top \tilde{C}$, and candidate residual $\tilde{u} = c^{(k)} - \bar{c}$ — one has

$$\text{rank}(\Sigma_{S \cup \{k\}}) > \text{rank}(\Sigma_S) \Leftrightarrow \tilde{u} \notin \text{range}(\tilde{C}^\top) = \text{range}(\tilde{\Sigma}_S) \Leftrightarrow (C_m^{(k)}, C_r^{(k)}) \notin \text{aff}(\mathcal{H}_S).$$

The general block form (3.1) replaces $\tilde{c}^{(s)} \tilde{c}^{(s)\top}$ by its residualized analogue; the row-space argument is identical.

Proof Work in the centered coordinates of the statement, so $\Sigma_{S \cup \{k\}} = \tilde{\Sigma}_S + \tilde{u} \tilde{u}^\top = \tilde{C}^\top \tilde{C} + \tilde{u} \tilde{u}^\top$. For any x ,

$$x^\top (\tilde{C}^\top \tilde{C} + \tilde{u} \tilde{u}^\top) x = \|\tilde{C}x\|_2^2 + (\tilde{u}^\top x)^2 \geq 0,$$

and the right-hand side is zero iff $\tilde{C}x = 0$ and $\tilde{u}^\top x = 0$. Since $\text{null}(\tilde{C}^\top \tilde{C}) = \text{null}(\tilde{C})$, this gives the identity

$$\text{null}(\tilde{C}^\top \tilde{C} + \tilde{u} \tilde{u}^\top) = \text{null}(\tilde{C}) \cap \{x : \tilde{u}^\top x = 0\}. \quad (\text{A.1})$$

The intersection is a proper subspace of $\text{null}(\tilde{C})$ iff the hyperplane $\{\tilde{u}^\top x = 0\}$ does not contain $\text{null}(\tilde{C})$, i.e. iff $\tilde{u} \notin \text{null}(\tilde{C})^\perp = \text{range}(\tilde{C}^\top) = \text{range}(\tilde{\Sigma}_S)$. By rank-nullity this is exactly a strict increase of rank, proving the first equivalence. For the second, the centered row space $\text{range}(\tilde{C}^\top) = \text{span}\{\tilde{c}^{(s)}\}$ is the linear subspace parallel to $\text{aff}(\mathcal{H}_S)$, and $\tilde{u}$ lies in it iff $(C_m^{(k)}, C_r^{(k)}) \in \text{aff}(\mathcal{H}_S)$. $\square$

Remark (Why centering is essential). The equivalence with affine-hull membership fails for the *uncentered* Gram matrix $C^\top C$, whose range is the linear span $\text{span}\{c^{(s)}\}$, not the direction space of the affine hull. For $c^{(1)} = (1, 0)$, $c^{(2)} = (0, 1)$ the affine hull is the line $\{x_1 + x_2 = 1\}$ while $\text{span}\{c^{(1)}, c^{(2)}\} = \mathbb{R}^2$; the candidate $u = (2, 2)$ lies outside the hull yet $C^\top C = I_2$ is already full rank, so $C^\top C + uu^\top$ does not increase rank. The affine-hull reading of Proposition 2 is therefore a statement about the *centered* (conditional-covariance) moment of (3.1), not the raw second-moment matrix.

A.3 Proof of Proposition 3 (width = distance $\times 2\tau$)

Claim. We prove Proposition 3 and Corollary 2: with θ_0 the minimum-norm solution of $\Sigma_S \theta = -b$ and $\delta_S(t) = \|P_{\text{null}(\Sigma_S)} c(t)\|_2$, for every $\tau \geq \|\theta_0\|_2$,

$$w_r^\tau(t; \Sigma_S, b) = 2\sqrt{\tau^2 - \|\theta_0\|_2^2} \delta_S(t) = 2\tau \delta_S(t)(1 + o_\tau(1)), \quad \tau \rightarrow \infty,$$

and, under the balanced embedding of Corollary 2, $\delta_S(t) = \sqrt{\kappa} \text{dist}((t_m, t_r), \text{aff}(\mathcal{H}_S))$.

Proof Decompose $c(t) = c_\parallel(t) + c_\perp(t)$ with $c_\parallel(t) \in \text{range}(\Sigma_S^\top)$ and $c_\perp(t) \in \text{null}(\Sigma_S)$, so $\|c_\perp(t)\|_2 = \delta_S(t)$. Take θ_0 the minimum-norm particular solution, so $\theta_0 \in \text{range}(\Sigma_S^\top)$ and $\theta_0 \perp \text{null}(\Sigma_S)$. Every feasible point is $\theta = \theta_0 + \nu$, $\nu \in \text{null}(\Sigma_S)$, with $c(t)^\top \theta = c_\parallel(t)^\top \theta_0 + c_\perp(t)^\top \nu$ since $c_\parallel(t)^\top \nu = 0$ and $c_\perp(t)^\top \theta_0 = 0$.

Because $\theta_0 \perp \nu$, the ball constraint reads $\|\theta\|_2^2 = \|\theta_0\|_2^2 + \|\nu\|_2^2 \leq \tau^2$, i.e. $\|\nu\|_2 \leq \sqrt{\tau^2 - \|\theta_0\|_2^2} =: \rho_\tau$. This is the *exact* radius, not $\tau - \|\theta_0\|_2$. Maximizing and minimizing the linear functional $c_\perp(t)^\top \nu$ over the radius- ρ_τ ball in $\text{null}(\Sigma_S)$ gives $\pm \rho_\tau \|c_\perp(t)\|_2$ by Cauchy–Schwarz, attained at $\nu = \pm \rho_\tau c_\perp(t) / \|c_\perp(t)\|_2$. Hence

$$w_r^\tau(t; \Sigma_S, b) = 2\rho_\tau \|c_\perp(t)\|_2 = 2\sqrt{\tau^2 - \|\theta_0\|_2^2} \delta_S(t) = 2\tau \delta_S(t)(1 + O(\tau^{-2})),$$

which is (4.3). The dominance claim follows since $\delta_{S \cup \{k\}}(t) \leq \delta_S(t)$: the PSD update only enlarges the range, $\text{range}(\Sigma_{S \cup \{k\}}^\top) \supseteq \text{range}(\Sigma_S^\top)$, so the distance to it cannot grow, with equality iff $c(t)$'s null-space component is unchanged — by Proposition 2, iff $\tilde{u} \in \text{range}(\tilde{\Sigma}_S)$.

For Corollary 2, write $x := (t_m, t_r) - \bar{c} = x_V + x_\perp$ with $x_V \in V := \text{span}\{\tilde{c}^{(s)}\}$ and $x_\perp \perp V$. The assumed range identity $\text{range}(\Sigma_S^\top) = LV$ is what licenses the pullback — without it, the distance from $c(t)$ to $\text{range}(\Sigma_S^\top)$ need not correspond to any distance in context space. Given it, $Lx_V \in LV = \text{range}(\Sigma_S^\top)$, and $L^\top L = \kappa I$ makes $Lx_\perp \perp LV$ (since $(Lx_\perp)^\top (Lv) = \kappa x_\perp^\top v = 0$), so $\delta_S(t) = \|Lx_\perp\|_2 = \sqrt{\kappa} \|x_\perp\|_2 = \sqrt{\kappa} \text{dist}((t_m, t_r), \text{aff}(\mathcal{H}_S))$, as claimed. V is the direction space of $\text{aff}(\mathcal{H}_S)$. $\square$

A.4 Proof of Theorem 5 (monotone improvement under (M1) or (M2))

We first establish the algebraic identity that drives the proof. Let $u := (u_m^{(k)}, u_r^{(k)})$ be the residualized context of the candidate and y_k its scalar outcome forecast. Then, in the simplified moment formulation $\Sigma_{\mathcal{S} \cup \{k\}} = \Sigma_{\mathcal{S}} + uu^\top$ and $b_{\mathcal{S} \cup \{k\}} = b_{\mathcal{S}} - uy_k$, we have for every θ :

$$\Sigma_{\mathcal{S} \cup \{k\}}\theta + b_{\mathcal{S} \cup \{k\}} = \Sigma_{\mathcal{S}}\theta + b_{\mathcal{S}} + u \cdot (u^\top\theta - y_k). \quad (\text{A.2})$$

Direct verification: $\Sigma_{\mathcal{S} \cup \{k\}}\theta = \Sigma_{\mathcal{S}}\theta + u(u^\top\theta)$ and $b_{\mathcal{S} \cup \{k\}} = b_{\mathcal{S}} - uy_k$; summing gives (A.2).

Proof Recall the identity (A.2): $\Sigma_{\mathcal{S} \cup \{k\}}\theta + b_{\mathcal{S} \cup \{k\}} = \Sigma_{\mathcal{S}}\theta + b_{\mathcal{S}} + u(u^\top\theta - y_k)$.

(ii) *Rank-expanding case*, $u \notin \text{range}(\Sigma_{\mathcal{S}})$. Let $P^\perp$ project orthogonally onto $\text{null}(\Sigma_{\mathcal{S}})$, so $u_\perp := P^\perp u \neq 0$. Applying $P^\perp$ to (A.2) and using $b_{\mathcal{S}} \in \text{range}(\Sigma_{\mathcal{S}})$ (which kills $\Sigma_{\mathcal{S}}\theta + b_{\mathcal{S}}$) gives $P^\perp[\Sigma_{\mathcal{S} \cup \{k\}}\theta + b_{\mathcal{S} \cup \{k\}}] = u_\perp(u^\top\theta - y_k)$. Hence $\theta \in \mathcal{L}(\Sigma_{\mathcal{S} \cup \{k\}}, b_{\mathcal{S} \cup \{k\}})$ forces $u^\top\theta = y_k$ (as $u_\perp \neq 0$), and then (A.2) forces $\Sigma_{\mathcal{S}}\theta + b_{\mathcal{S}} = 0$; conversely any $\theta \in \mathcal{L}(\Sigma_{\mathcal{S}}, b_{\mathcal{S}})$ with $u^\top\theta = y_k$ satisfies $\Sigma_{\mathcal{S} \cup \{k\}}\theta + b_{\mathcal{S} \cup \{k\}} = 0$. Intersecting with $\mathcal{R}$,

$$\mathcal{F}_{\mathcal{S} \cup \{k\}} = \mathcal{F}_{\mathcal{S}} \cap \{\theta : u^\top\theta = y_k\}.$$

If this set is non-empty it is contained in $\mathcal{F}_{\mathcal{S}}$, so for every t the supremum (resp. infimum) of $c(t)^\top\theta$ over it is no larger (resp. no smaller), giving the pointwise width inequality. Under (M2), $y_k = u^\top\theta_{\mathcal{S}}^*$, the hyperplane passes through $\theta_{\mathcal{S}}^* \in \mathcal{L}(\Sigma_{\mathcal{S}}, b_{\mathcal{S}})$, which lies in $\mathcal{F}_{\mathcal{S}}$ as soon as $\theta_{\mathcal{S}}^* \in \mathcal{R}$; then $\theta_{\mathcal{S}}^* \in \mathcal{F}_{\mathcal{S} \cup \{k\}}$ and non-emptiness holds. Absent such a guarantee the augmented set may be empty (see the Remark below), and the inequality is vacuous.

(i) *Rank-preserving case*, $u \in \text{range}(\Sigma_{\mathcal{S}})$, (M2). Here $u \perp \text{null}(\Sigma_{\mathcal{S}})$. For $\theta \in \mathcal{F}_{\mathcal{S}}$, write $\theta = \theta_{\mathcal{S}}^* + \nu$, $\nu \in \text{null}(\Sigma_{\mathcal{S}})$; then $u^\top\theta - y_k = u^\top\nu = 0$, so by (A.2) $\theta \in \mathcal{L}(\Sigma_{\mathcal{S} \cup \{k\}}, b_{\mathcal{S} \cup \{k\}})$. The two affine subspaces share $\theta_{\mathcal{S}}^*$ (by (M2)) and the same direction subspace — by (A.1), $\text{null}(\Sigma_{\mathcal{S}} + uu^\top) = \text{null}(\Sigma_{\mathcal{S}}) \cap \{u^\top x = 0\} = \text{null}(\Sigma_{\mathcal{S}})$ since $u \perp \text{null}(\Sigma_{\mathcal{S}})$ — hence coincide. Intersecting with $\mathcal{R}$ gives $\mathcal{F}_{\mathcal{S} \cup \{k\}} = \mathcal{F}_{\mathcal{S}}$, so the width is unchanged target by target. Taking suprema over $t \in \mathcal{T}$ in either case proves the worst-case statement. $\square$

Remark (Feasibility is needed: (M1) alone can break it). The hypothesis $\mathcal{F}_{\mathcal{S} \cup \{k\}} \neq \emptyset$ in (ii) is not implied by (M1) and consistency. Take $\Sigma_{\mathcal{S}} = \text{diag}(1, 0)$, $b_{\mathcal{S}} = 0$, $\mathcal{R} = [-1, 1]^2$, $u = (0, 1)$, $y_k = 100$. Then $u \notin \text{range}(\Sigma_{\mathcal{S}})$ (so (M1) holds) and $b_{\mathcal{S}} \in \text{range}(\Sigma_{\mathcal{S}})$, with $\mathcal{F}_{\mathcal{S}} = \{0\} \times [-1, 1] \neq \emptyset$. But $\Sigma_{\mathcal{S} \cup \{k\}} = I$ and $b_{\mathcal{S} \cup \{k\}} = (0, -100)$ force $\theta = (0, 100) \notin \mathcal{R}$, so $\mathcal{F}_{\mathcal{S} \cup \{k\}} = \emptyset$ and the augmented interval is empty (failure mode (F3)). This is why (ii) is conditional on non-emptiness, and why the safe screen pairs (M1) with the outcome control (M2)/ $\theta_{\mathcal{S}}^* \in \mathcal{R}$.

A.5 Proof of Proposition 4 (rank-2 envelope)

Proposition 4 is a special case of the Bonnans–Shapiro directional derivative theorem for the value function of a perturbed convex program. We sketch the specialization to (2.3) to make the constants explicit.

Setup. For the upper endpoint define

$$v_u(h) := \max_{\theta} \{c(t)^\top\theta : (\Sigma_{\mathcal{S}} + h_{\Sigma})\theta = -(b_{\mathcal{S}} + h_b), r(\theta) \leq 0\},$$

with perturbation $h = (h_{\Sigma}, h_b)$ and KKT triple $(\theta_u^*, \lambda_u^*, \nu_u^*)$ at $h = 0$; stationarity reads $c(t) = \Sigma_{\mathcal{S}}\lambda_u^* + \partial r(\theta_u^*)^\top \nu_u^*$. Define $v_\ell(h)$ symmetrically with the min Lagrangian and triple $(\theta_\ell^*, \lambda_\ell^*, \nu_\ell^*)$.

Directional derivative. Under Robinson’s constraint qualification (Assumption 6 of [Boyarsky et al., 2026](#)) and strict complementarity, each value function is directionally differentiable with

$$Dv_u \cdot h = -\lambda_u^{\star\top} (h_\Sigma \theta_u^\star + h_b), \quad Dv_\ell \cdot h = -\lambda_\ell^{\star\top} (h_\Sigma \theta_\ell^\star + h_b),$$

each evaluated at its *own* optimizer and multiplier (Theorem 4.24 of [Bonnans and Shapiro, 2013](#)). Without strict complementarity the derivatives are Gol’shtein-type minimax values over the optimal primal–dual sets: for the upper (max) problem the multiplier is selected by an inner *minimum* (its value function is concave in the right-hand side), and for the lower (min) problem by an inner *maximum* (convex value function), as stated in Proposition 4.

Application to site augmentation. Set $h_\Sigma = \Delta \Sigma_k$, $h_b = \Delta b_k$ to obtain (5.1); subtracting, $Dw_k = Dv_u - Dv_\ell$. The endpoints share an optimizer — so that $Dw_k = -(\lambda_u^\star - \lambda_\ell^\star)^\top (\Delta \Sigma_k \theta^\star + \Delta b_k)$ — only in the degenerate zero-width case $\theta_u^\star = \theta_\ell^\star$; in general the optimizers, active sets, and multipliers differ and are kept separate.

Rank-2 reduction. By (3.1), $\Delta \Sigma_k = U_k A_k U_k^\top$ with U_k a $d_g \times 2$ factor, so $\Delta \Sigma_k \theta_\bullet^\star = U_k (A_k U_k^\top \theta_\bullet^\star)$ lies in the two-dimensional column space of U_k , and each endpoint derivative reduces to the two scalars $\lambda_\bullet^{\star\top} U_k$. The minimax (3.2) over $k \in \mathcal{K}$ therefore reduces to evaluating two scalars per candidate after the existing-pool LPs have been solved once; this is the $O(|\mathcal{T}|K)$ cost claimed in Section 5.1.

A.6 Extension to the block (BEN) moment

The proofs of Appendices A.1, A.2, and A.4 use only three facts: Σ is symmetric PSD, the update $\Delta \Sigma_k$ is PSD of rank at most two, and the algebraic identity (A.2). We record the block form that supplies the first two.

With $w := (u_m, u_r) \in \mathbb{R}^{d_M+d_I}$ and $ww^\top$ the corresponding $(d_M+d_I) \times (d_M+d_I)$ outer product, the update implied by Equation (A.9) of [Boyarsky et al. \[2026\]](#) is

$$\Delta \Sigma_k = n_k \rho \begin{bmatrix} ww^\top & ww^\top \\ ww^\top & \rho^{-1} ww^\top \end{bmatrix} = U_k A_k U_k^\top, \quad U_k = \begin{bmatrix} w & 0 \\ 0 & w \end{bmatrix}, \quad A_k = n_k \rho \begin{bmatrix} 1 & 1 \\ 1 & \rho^{-1} \end{bmatrix}, \quad (\text{A.3})$$

where the four row/column blocks have sizes (d_M, d_I, d_M, d_I) to match $\theta = (\beta_1, \beta_2, \gamma_1, \gamma_2)$, and each displayed $ww^\top$ is the symmetric block $\begin{bmatrix} u_m u_m^\top & u_m u_r^\top \\ u_r u_m^\top & u_r u_r^\top \end{bmatrix}$. (Two corrections to the earlier display: its top-right and bottom-left blocks had the off-diagonal sub-blocks transposed — dimensionally consistent only when $d_M = d_I$ — and the prefactor is $+n_k \rho$, not $n_k(-\rho)$.) For a constant propensity $\rho \in (0, 1]$ the 2×2 core has trace $1 + \rho^{-1} > 0$ and determinant $\rho^{-1} - 1 \geq 0$, so $A_k \succeq 0$; hence $\Delta \Sigma_k \succeq 0$, $\Sigma_{\mathcal{S} \cup \{k\}} \succeq \Sigma_{\mathcal{S}}$, and $\text{rank}(\Delta \Sigma_k) \leq \text{rank}(U_k) \leq 2$.

Given (A.3), the identity (A.2) holds in the block model with the rank-one term $u(u^\top \theta - y_k)$ replaced by $U_k A_k (U_k^\top \theta - \eta_k)$, where η_k is the block-residualized outcome moment: for the rank- r update $\Sigma' = \Sigma + U_k A_k U_k^\top$, $b' = b - U_k A_k \eta_k$,

$$\Sigma' \theta + b' = \Sigma \theta + b + U_k A_k (U_k^\top \theta - \eta_k). \quad (\text{A.4})$$

The rank-one monotonicity proof extends under the sufficient condition

$$\text{range}(U_k A_k) \cap \text{range}(\Sigma_{\mathcal{S}}) = \{0\}, \quad \text{equivalently} \quad \ker(P_{\text{null}(\Sigma_{\mathcal{S}})} U_k A_k) \subseteq \ker(U_k A_k). \quad (\text{A.5})$$

Indeed, for $\theta \in \mathcal{L}(\Sigma', b')$, projecting (A.4) onto $\text{null}(\Sigma_{\mathcal{S}})$ (using $b_{\mathcal{S}} \in \text{range}(\Sigma_{\mathcal{S}})$) gives $P_{\text{null}(\Sigma_{\mathcal{S}})} U_k A_k (U_k^\top \theta - \eta_k) = 0$, and (A.5) upgrades this to $U_k A_k (U_k^\top \theta - \eta_k) = 0$, hence $\Sigma_{\mathcal{S}} \theta + b_{\mathcal{S}} = 0$. Therefore

$$\mathcal{F}_{\mathcal{S} \cup \{k\}} = \mathcal{F}_{\mathcal{S}} \cap \{\theta : U_k A_k (U_k^\top \theta - \eta_k) = 0\},$$

and, whenever this intersection is non-empty, every support-function width weakly decreases, exactly as in Theorem 5(ii). We emphasize that (A.5) is a condition on the *whole update range*: requiring merely that each column of U_k lie outside $\text{range}(\Sigma_S)$ is *not* sufficient, since a linear combination of the columns can fall inside $\text{range}(\Sigma_S)$ (e.g. columns $e_1 \pm e_2$ with $\text{range}(\Sigma_S) = \text{span}(e_1)$), and then the projection step above no longer pins down the old moment. If the update has a component inside $\text{range}(\Sigma_S)$, that component requires an (M2)-type consistency condition on the matching coordinates of η_k ; otherwise it translates the old affine slice and monotonicity can fail as in Section 6.2. Pinning down the exact constants n_k, ρ from the BEN estimator is left to their Appendix A.9; nothing in Sections 4–6 uses more than the PSD, rank- ≤ 2 , identity, and transversality properties verified here.

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